



CC6a
Covalent Bonding

Covalent Bonds

03 October 2024

Recap activity: Explain why ionic compounds do not conduct electricity when solid

Learning Objective: To how understand covalent bonds formed.

Developing Recall the names of some common molecular elements and compounds

Securing Recall that covalent bonding results in the formation of molecules

Exceeding Explain the formation of simple molecular, covalent substances, using dot and cross diagrams

Extending Explain the formation of simple molecular, covalent substances, using dot and cross diagrams

Learning Outcomes:

CC1.28 Explain how covalent bonds are formed.

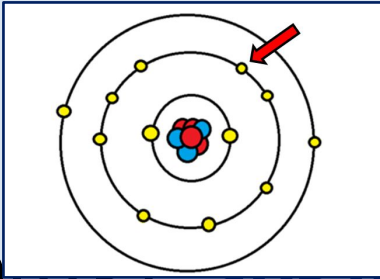
CC1.29 Recall the names of some common molecular elements.

CC1.29 Recall the names of some common molecular compounds.

CC1.29 State the bonding that is found in molecules.

CC1.30 Explain the formation of covalent bonds using dot and cross diagrams.

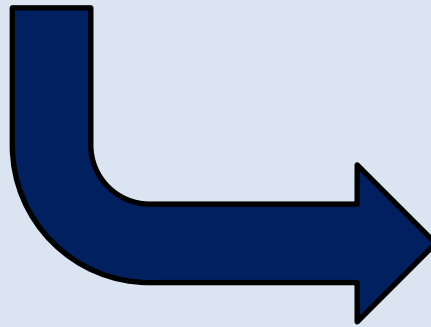
Covalent Bonds

Covalent  usually occur between  atoms and are produced through sharing  electrons to complete their outer shell.  

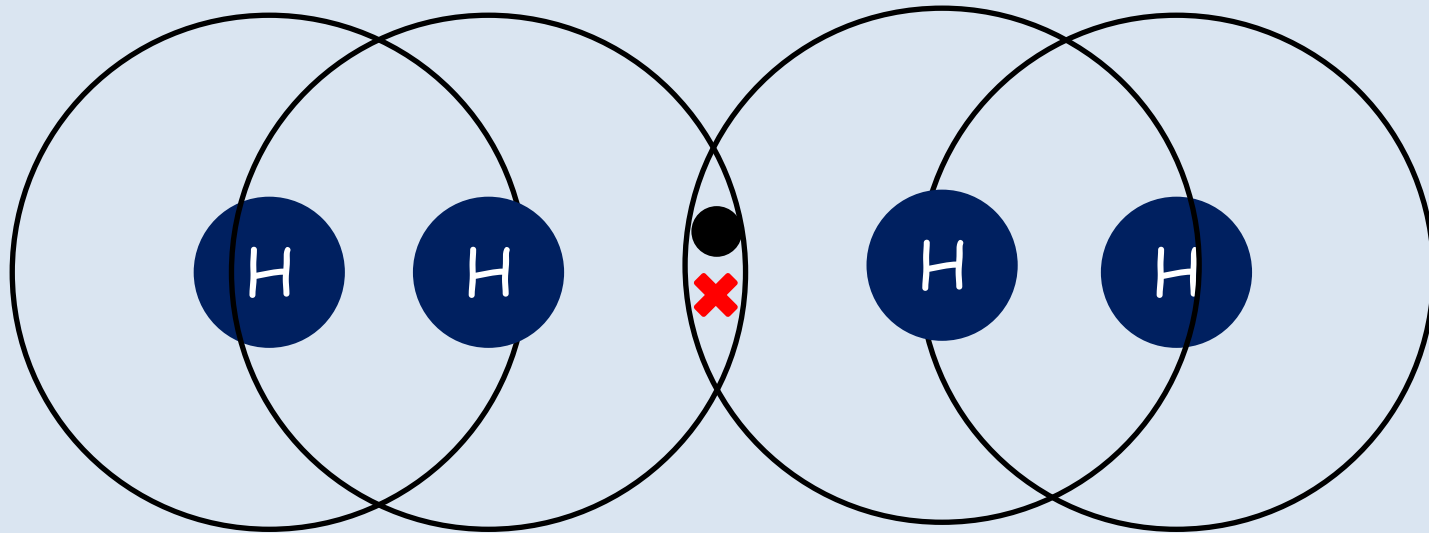
Extending: State one similarity and one difference between covalent bonds and ionic bonds.

Covalent Bonds

Covalent bonds usually occur between non-metal **atoms** and are produced through sharing pairs of **electrons** to complete their **outer shell**.



Example: Hydrogen

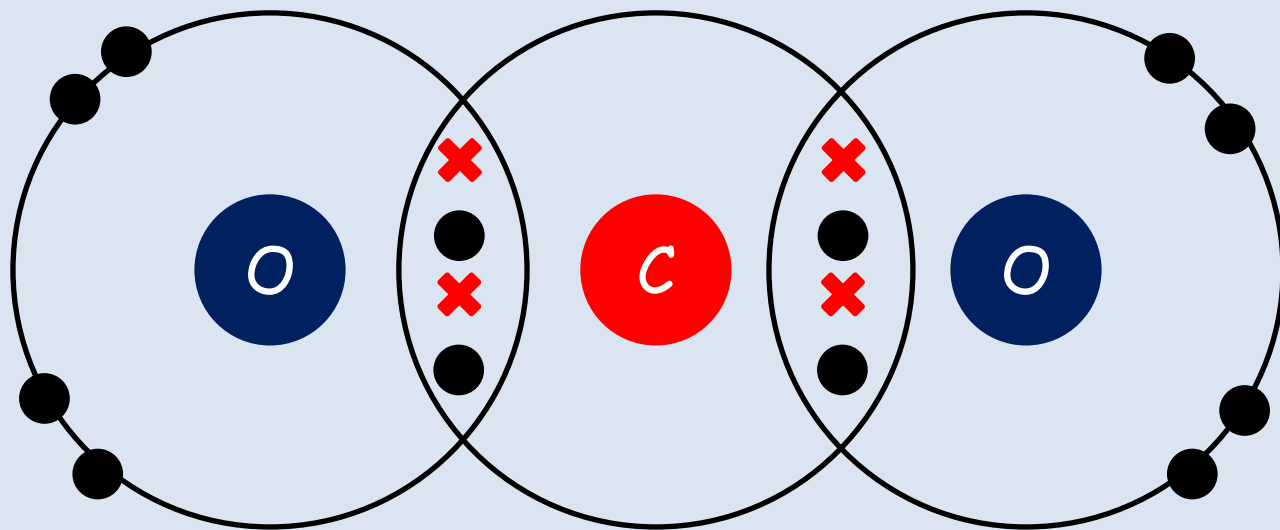


Hydrogen has one **electron** in it's **outer shell**.

Hydrogen **molecules** are formed by sharing electrons.

Two or more atoms joined by covalent bonds.

Example: Carbon Dioxide



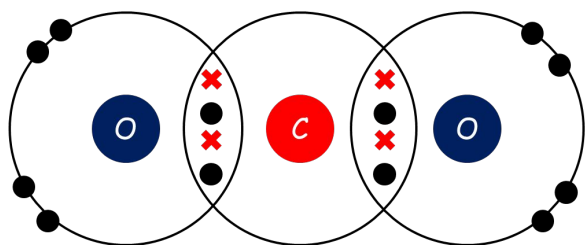
*(Only the
outer shell is
shown)*

Carbon has 4 **electrons** in it's **outer shell** (it needs 4 more to become **stable**).

Oxygen has 6 electrons in it's outer shell (it needs 2 more to become stable).

So 2 oxygens **share** their electrons with a carbon forming a **double covalent bond**.

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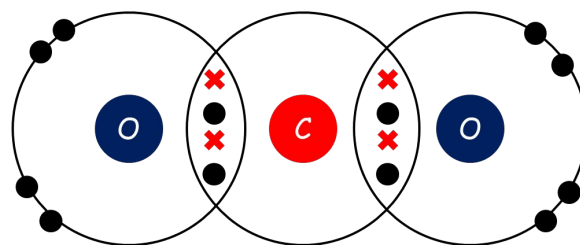


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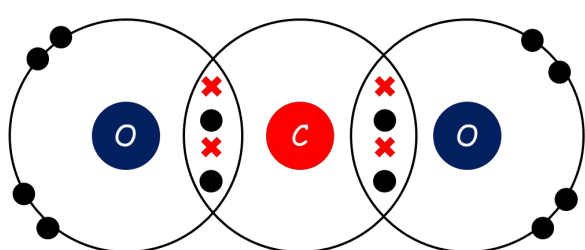


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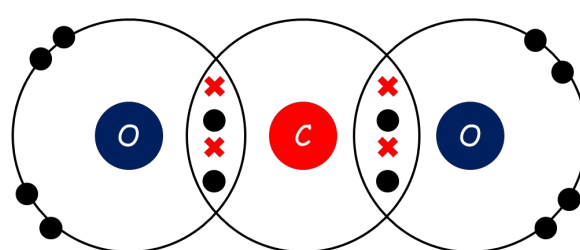


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Covalent Bonds

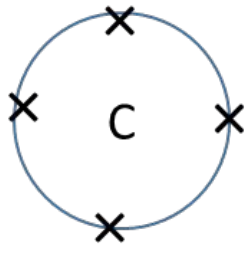
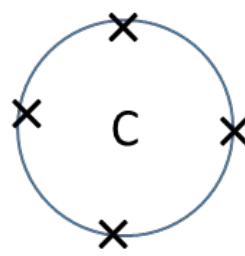
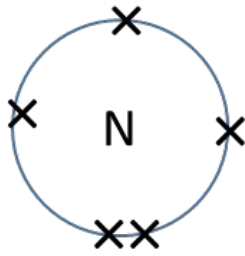
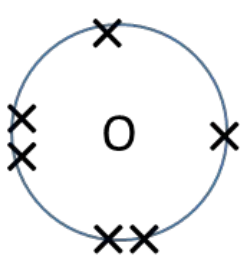
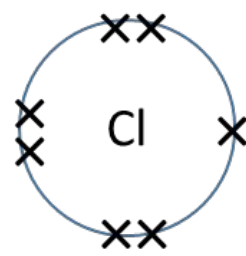
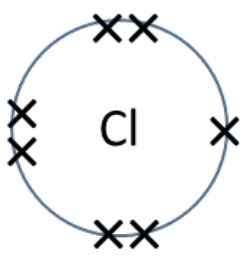
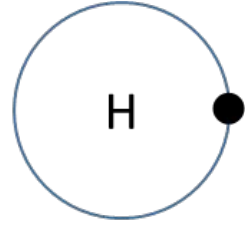
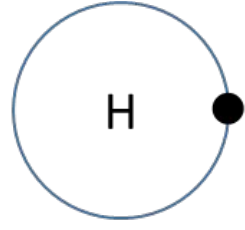
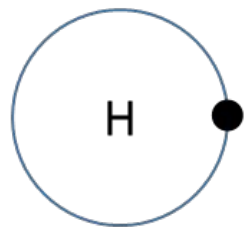
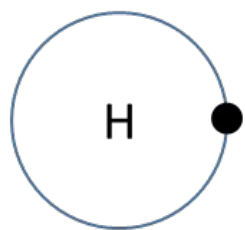
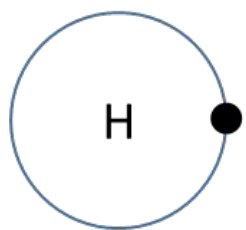
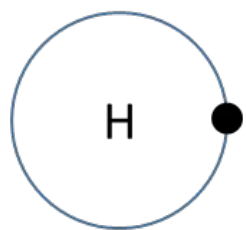
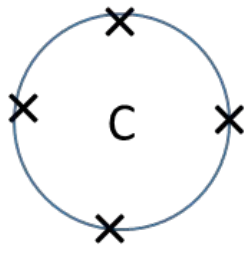
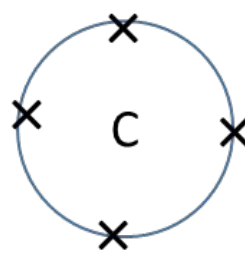
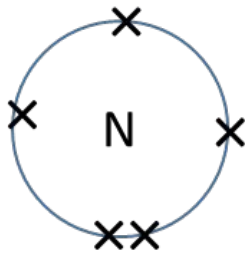
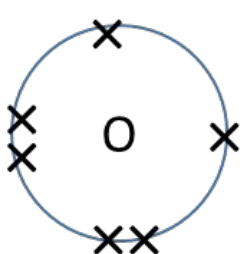
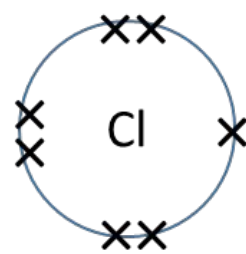
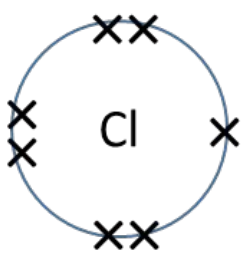
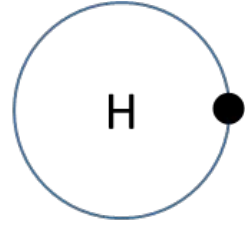
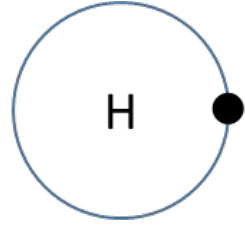
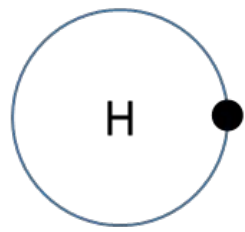
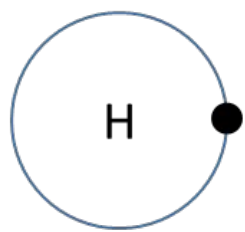
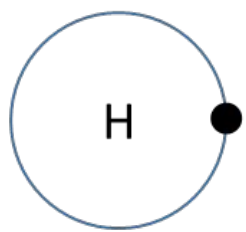
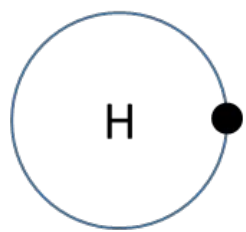
Hydrogen, H_2

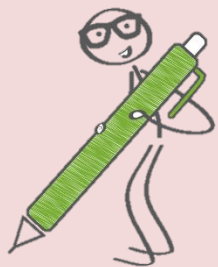
Water, H_2O

Hydrogen Chloride

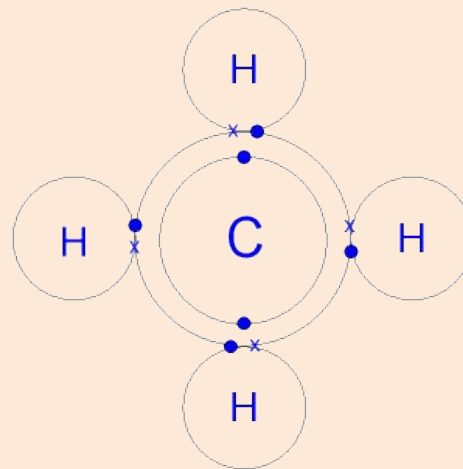
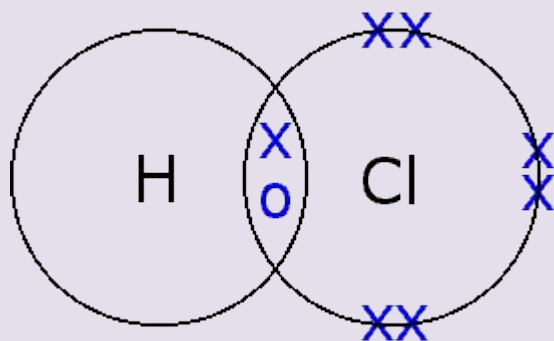
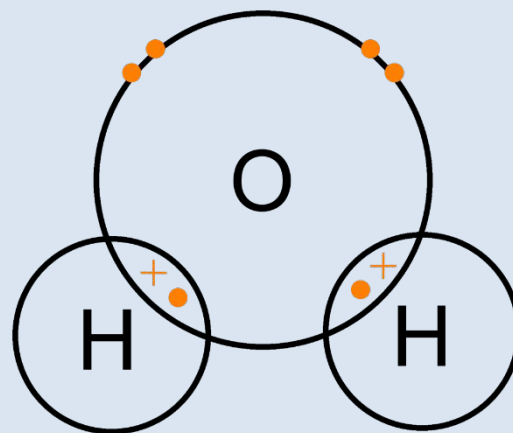
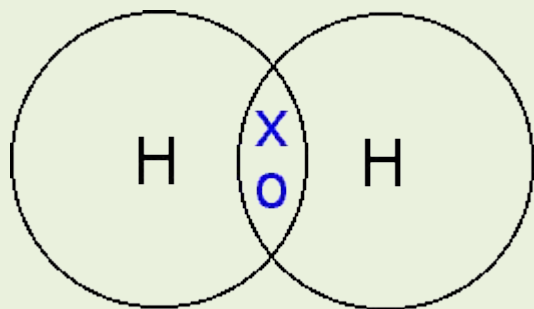
Methane, CH_4

CHALLENGE: State one similarity and one difference between covalent bonds and ionic bonds.





Answers:

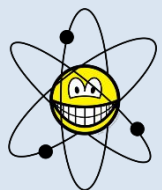


Electrostatic Forces

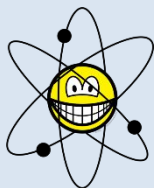
The **atoms** in **molecules** are held together by **strong electrostatic forces of attraction** between the **positive nuclei** and the **negative electrons** in the bonded atoms.



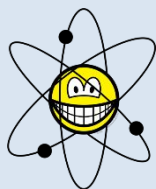
Progress Check



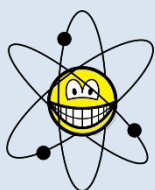
State what a molecule is.



Describe how a covalent bond is formed.



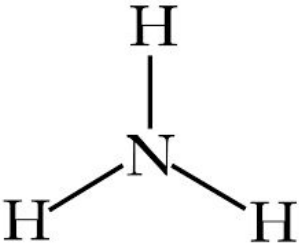
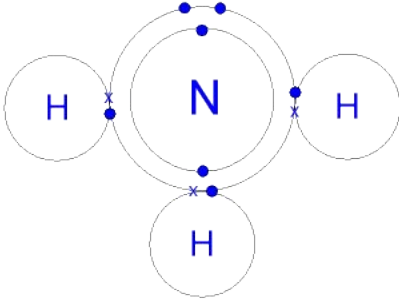
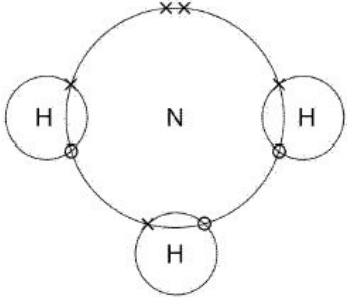
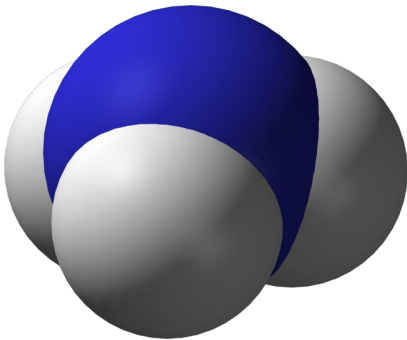
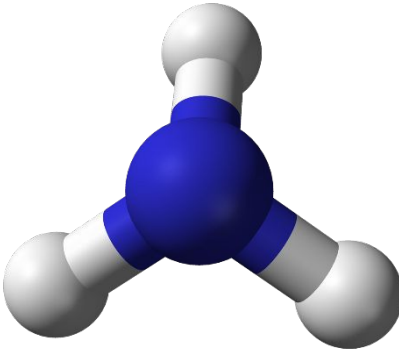
Chlorine has the configuration (2,8,1). Draw a dot and cross diagram showing how Cl_2 is formed.



Nitrogen (2,8,5) forms a triple bond with other nitrogen atoms. Draw a dot and cross diagram for N_2 .

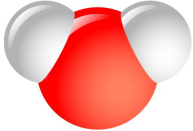
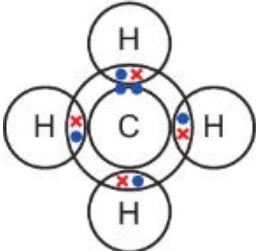
Representing Covalent Bonds

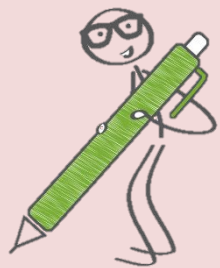
Ammonia:

Molecular Formula NH_3	Structural Formula 	Full Dot and Cross Diagram 
Dot and Cross Diagram (outer shell only) 	3D Space Filling 	Ball and Stick 

Representing Covalent Bonds

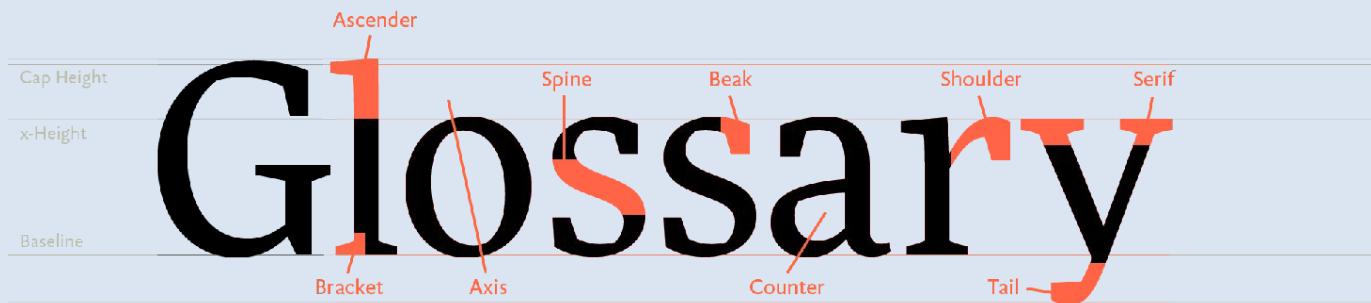
Turn your book sideways and draw the following table. Fill in the missing boxes.

Molecule	Formula	Structural	Full Dot and Cross	Outer Shell Dot and Cross	3D	Ball and Stick
	H ₂ O					
Methane						



Answers:

Molecule	Formula	Structural	Full Dot and Cross	Outer Shell Dot and Cross	3D	Ball and Stick
Water	H_2O	$H-O-H$				
Methane	CH_4	$\begin{array}{c} H \\ \\ H-C-H \\ \\ H \end{array}$				



Copy down the following key words and their definitions:

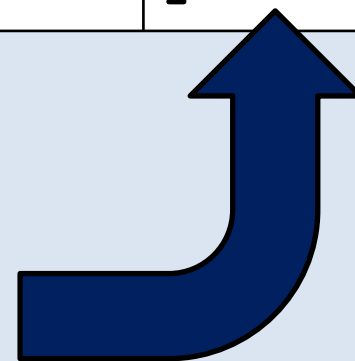
Valency- The number of covalent bonds formed by an atom. It is the same as the number of electrons needed to obtain a complete outer shell.

CHALLENGE: Suggest how an element's group number in the periodic table is linked to its valency.

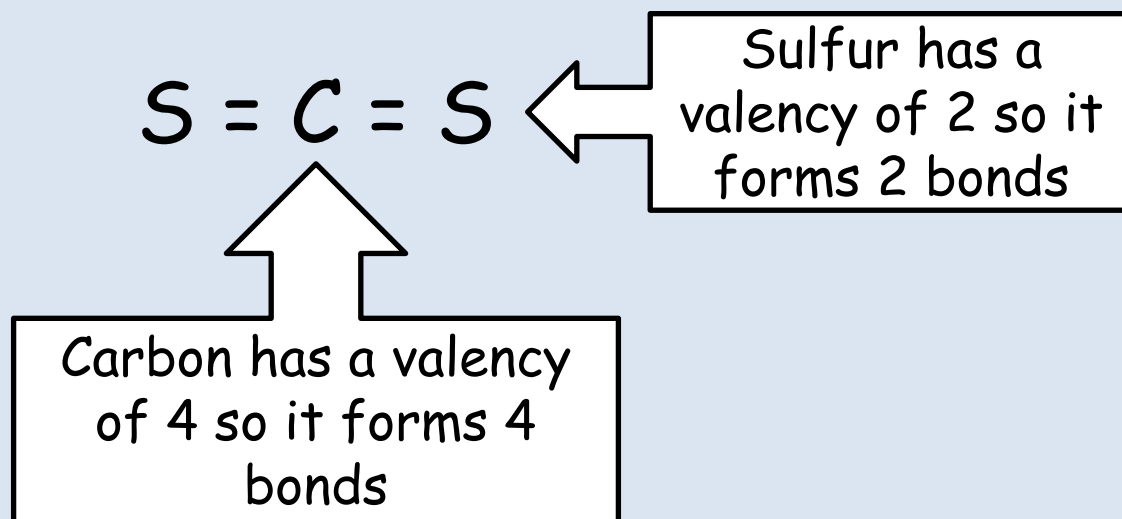
Valency

Group Number	Examples	Outer Electrons	Bonds Formed	Valency
4	C and Si	4	4	4
5	N and P	5	3	3
6	O and S	6	2	2
7	F and Cl	7	1	1

We can work out molecular formulae by matching up the valencies (making sure they have the correct number of bonds to complete their outer electron shell).



Molecular Formulae



So two **S atoms** each form a **double bond** with a single **C atom**. Giving each atom the correct number of **bonds**.

Therefore the molecular formula is **CS_2**

Molecular Formulae

Work out the following molecular formulae:

- Oxygen fluoride OF_2
- Hydrogen sulfide H_2S
- Nitrogen chloride NCl_3
- Carbon chloride CCl_4

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Retrieval Practice

How many points can you get in the time limit?

1 point- this lesson

2 points- last lesson

3 points- last topic

4 points- last year

State the molecular formula for a compound made of Iodine and Nitrogen.

Describe how an elements electron configuration is linked to its position in the periodic table.

Describe the particle arrangement and movement in a liquid.

Describe the different ways you can represent covalently bonded molecules.

Explain why ionic compounds can conduct electricity when dissolved in water.

Define the term 'mixture'.

State the properties of ionic compounds.

Explain how Mendeleev created his original periodic table.

State what an elements atomic number tells us.

Draw a dot and cross diagram to show how fluorine molecules are formed.

Explain how you can separate substances by crystallisation.

Describe how ionic compounds are bonded together.